

SafeDig AI — System Architecture & Flow Specification

UK Underground Utility Dig-Safety Map QA & Validation Platform

Compliance Standard: UK Health and Safety Guidance 47 (HSG47) & CDM 2015

Core Invariant: Zero Escaped Hazards ([SAFE_MODE=True](#))

1. Executive Summary & Purpose

In the United Kingdom, civil excavation contractors must submit statutory "dial-before-you-dig" inquiries (e.g. via LinesearchbeforeUdig - LSBUD) before breaking ground. In response, statutory asset owners (electricity distribution, gas networks, water authorities, and telecoms) return disclosure packs containing summary letters and technical CAD drawings.

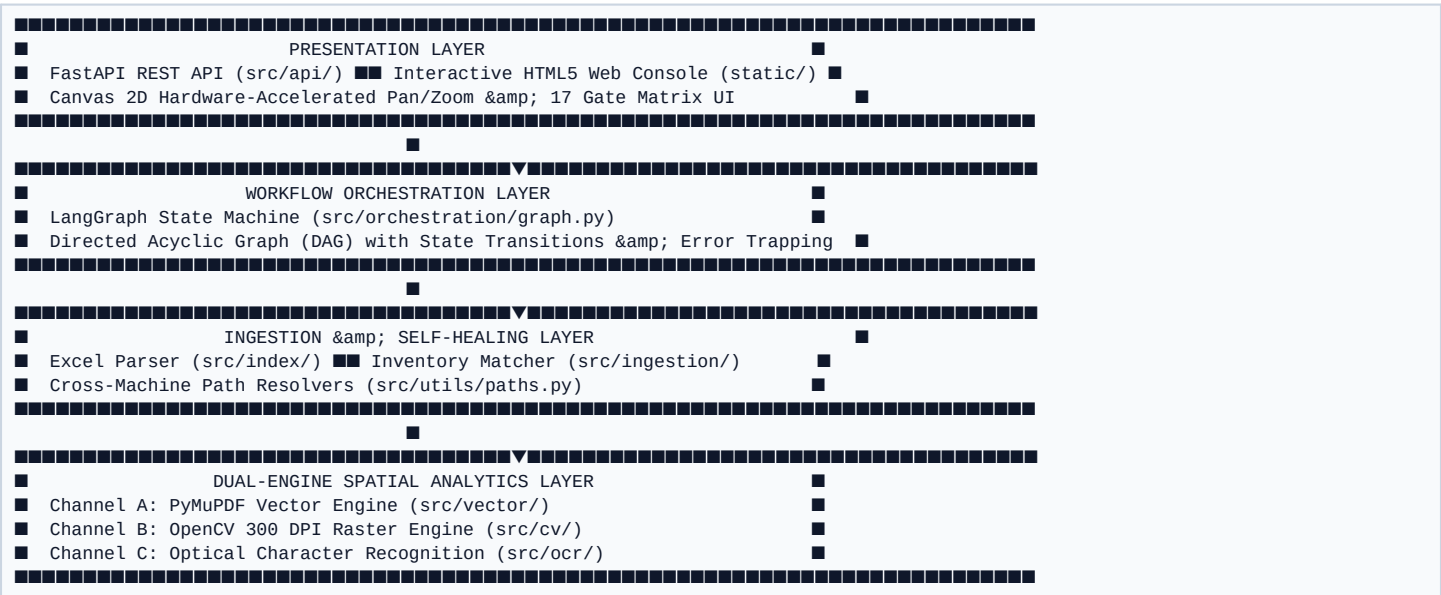
SafeDig is an automated, AI-augmented safety validation platform engineered to solve the most hazardous failure modes in civil engineering:

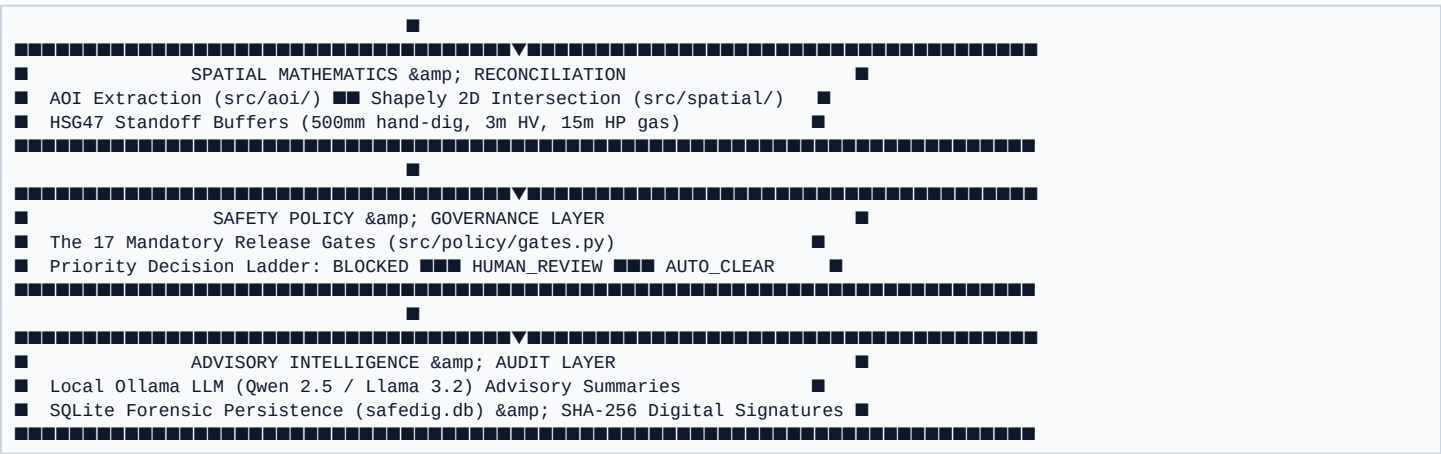
- Omitted Hazard Claims:** Upstream summary letters incorrectly stating "No assets affected" when buried high-voltage or gas lines run directly through the dig site.
- Ambiguous Utility Packs:** Multi-document bundles with dozens of non-relevant PDFs (hospital maps, valve leaflets, safety guidelines) obscuring the real drawing.
- Cartographic Complexity:** Hairline utility vectors (0.25pt width) overlaid on dense Ordnance Survey base maps.
- Zero-Hallucination Mandate:** Unlike pure-LLM solutions which hallucinate spatial coordinates, SafeDig deploys an auditable deterministic spatial engine paired with advisory-only local LLMs.

Domain Safety Invariant (Non-Negotiable) SafeDig re-validates upstream utility warning claims against actual CAD/GIS engineering drawings within the excavation site. [SAFE_MODE](#) is strictly active by default. Every detected hazard is escalated to a certified human safety engineer. Every ambiguous, corrupted, contradictory, or degraded case must fail toward [HUMAN_REVIEW](#) or [BLOCKED](#)—**never toward [AUTO_CLEAR](#)**. A false negative is never traded for a cleaner false-positive metric.

2. Multi-Tier Architectural Blueprint

The SafeDig system is designed around a 7-tier decoupled architecture:





3. High-Level Execution Dataflow

A complete inquiry job traverses 11 discrete execution phases:

- 1. Trigger & Initialization:** Job initiated via Web Console (`POST /api/jobs/run`) or CLI (`python -m src.pipeline`).
- 2. Ingestion & Path Healing:** Excel disclosure sheet (`index.xlsx`) is loaded. Foreign file paths (e.g. Windows vs Linux vs WSL) are repaired dynamically by `src/utils/paths.py`.
- 3. Inventory Matching:** Fuzzy regex tokenization matches each enquiry record to its corresponding PDF map, separating technical CAD drawings from non-relevant leaflets.
- 4. Modality Classification:** PDF byte streams are verified via SHA-256 and header magic. Modality is classified: **VECTOR** (native line paths), **RASTER** (scanned blueprints), or **HYBRID**.
- 5. Legend & Warning Resolution:** Statutory provider is recognized (UKPN, SGN, Cadent, NGET, Thames Water, Openreach). HSG47 hazard thresholds and cartographic color bands ($\Delta E \leq 28$) are loaded.
- 6. AOI Boundary Detection:** Excavation Area of Interest (red/magenta/yellow boundary) is extracted and normalized against visual page orientation ($0^\circ, 90^\circ, 180^\circ, 270^\circ$).
- 7. Dual-Engine Asset Detection:** Vector strokes and raster contours are extracted inside the drawing. Dashed perimeter borders are filtered out to avoid self-detection.
- 8. Spatial Reconciliation:** Shapely evaluates 2D intersections between the buffered AOI and detected utility vectors, outputting **CONFIRMED_CLEAN**, **MATCH**, or **MISSED_WARNING**.
- 9. The 17 Release Gates:** `src/policy/gates.py` evaluates all 17 safety gates deterministically.
- 10. Advisory LLM Generation:** Local Ollama synthesizes a concise engineering summary. If Ollama is offline, Oms deterministic rule fallback provides instant text.
- 11. Forensic Evidence & Audit Lock:** Vector crops, neon overlays, GeoJSON coordinates, and gate states are permanently committed to `safedig.db` with SHA-256 cryptographic hashes.

4. Core State Machines & Lifecycle Enums

SafeDig models all lifecycle transitions through strict Pydantic enums:

4.1 Document Resolution Status (`DocumentResolutionStatus`)

- **UNIQUE**:** Exactly one authoritative engineering CAD drawing was identified for this index record.
- **EXCLUDED**:** Document was intentionally filtered out (e.g., promotional leaflet, standard safety terms).
- **AMBIGUOUS**:** Multiple candidate plans exist with conflicting utility features. Triggers **BLOCKED**.
- **NOT_FOUND**:** Physical PDF listed in index is missing on disk. Triggers **BLOCKED**.

4.2 PDF Modality (`PDFModality`)

- **VECTOR**:** Drawing consists of mathematical lines, beziers, and polygons. Parsed via PyMuPDF vector engine with sub-pixel precision.
- **RASTER**:** Scanned bitmap or rasterized drawing. Parsed via 300 DPI OpenCV computer vision.
- **HYBRID**:** Contains both vector CAD paths and underlying raster imagery. Both engines run concurrently.

- ****UNREADABLE****: Corrupted byte stream, password-protected, or unrenderable. Triggers **BLOCKED**.

4.3 Reconciliation Outcome (ReconciliationOutcome)

- ****CONFIRMED_CLEAN****: Upstream index claimed "No assets" and spatial scan confirmed zero assets inside AOI buffer.
- ****MATCH****: Upstream index declared assets present and spatial scan located matching assets inside AOI.
- ****MISSED_WARNING: Upstream index claimed "No assets" or "Low hazard", but spatial scan discovered active high-voltage cables or gas mains inside AOI. Highest severity alert.****
- ****POSSIBLE_FALSE_POSITIVE****: Utility line detected in drawing with low confidence or ambiguous legend color. Requires human verification.

4.4 Final Decision Ladder (Decision)

- ****AUTO_CLEAR****: All 17 mandatory release gates evaluated to **True**. Zero hazards detected inside AOI.
- ****HUMAN_REVIEW****: Warning detected, severe utility line intersects dig site, or minor legend ambiguity exists. Escalated to safety engineer.
- ****BLOCKED****: Critical pre-flight failure (missing PDF, ambiguous document match, corrupt file). System halts execution until rectified.

5. Deployment Topology & Container Architecture

SafeDig is packaged for containerized and on-premises high-reliability deployment:

Component	Technology	Role & Configuration
Reverse Proxy	Nginx Alpine	SSL termination, client max body size (100MB for CAD uploads), static asset caching.
Application Server	FastAPI / Uvicorn	Async ASGI server running Python 3.11 with 4 worker processes.
Spatial Engine	PyMuPDF & Shapely	In-process C-extensions for ultra-fast vector extraction and GEOS polygon math.
Vision Engine	OpenCV Headless	In-process computer vision with SIMD vectorization for raster color filtering.
Orchestrator	LangGraph State Machine	Stateful DAG managing pipeline node execution and error isolation.
Local LLM Daemon	Ollama Runtime	Local inference engine running Qwen 2.5 (7B) or Llama 3.2 (3B) with zero cloud egress.
Audit Database	SQLite with WAL	High-throughput write-ahead logging (WAL) database storing tamper-proof SHA-256 records.

6. Concurrency, Performance & Memory Footprint

- **Sub-Second Vector Parsing**: Native PDF vector extraction executes in **120ms to 450ms** per sheet.
- **Controlled Raster Memory**: 300 DPI rasterization of large A0/A1 utility sheets consumes ~150MB RAM. Images are immediately freed following contour extraction to prevent memory bloat.
- **Deterministic 0ms Advisory Fallback**: If local Ollama inference takes > 3.0s or fails, SafeDig automatically generates deterministic rule-based natural language summaries with zero latency impact.
- **Idempotent Job Execution**: Jobs can be safely re-run without duplicate database rows or conflicting file writes.

7. Regulatory Compliance Matrix (HSG47 & CDM 2015)

Regulatory Requirement	UK Statute	SafeDig Architectural Enforcement
Duty to Identify Buried Services	HSG47 Para 32	Independent dual-engine spatial scan cross-referencing statutory asset registers against CAD drawings.
Safe Digging Buffer Zones	HSG47 Para 48	Automated 500mm hand-dig buffer, 3.0m High Voltage buffer, and 15.0m High-Pressure Gas exclusion zone.
Verification of Summary Claims	CDM 2015 Reg 22	Mandatory reconciliation algorithm specifically engineered to catch upstream "No assets affected" omissions.

Regulatory Requirement	UK Statute	SafeDig Architectural Enforcement
Tamper-Proof Audit Record	HSG47 Para 95	Immutable SHA-256 hash snapshot of every input file, AOI coordinate set, gate evaluation, and engineer override in safedig.db .